- Group By clause
- Distinct keyword
- Columns contain by expressions
- NOT NULL columns in the base table that are not selected by the view

**Example: (Using the WITH CHECK OPTION clause)**

```
CREATE OR REPLACE VIEW empvu20
 AS    SELECT *
FROM employees
WHERE department_id=20
WITH CHECK OPTION CONSTRAINT empvu20_ck;
```

**Note:** Any attempt to change the department number for any row in the view fails because it violates the WITH CHECK OPTION constraint.

**Example** – (Execute this and note the error)

```
UPDATE empvu20 SET department_id=10 WHERE employee_id=201;
```

**Denying DML operations**

Use of WITH READ ONLY option.
Any attempt to perform a DML on any row in the view results in an oracle server error.

**Try this code:**

```
CREATE OR REPLACE VIEW empvu10(employee_number, employee_name,job_title)
AS SELECT employee_id, last_name, job_id
FROM employees
WHERE department_id=10
WITH READ ONLY;
```

**Find the Solution for the following:**

1. Create a view called EMPLOYEE_VU based on the employee numbers, employee names and department numbers from the EMPLOYEES table. Change the heading for the employee name to EMPLOYEE.

```
CREATE VIEW employee-VU AS
SELECT Employee-id, Employee-name AS EMPLOYEE, department-id
FROM Employees;
```

2. Display the contents of the EMPLOYEES_VU view.

```
SELECT * FROM EMPLOYEE_VU ;
```

3. Select the view name and text from the USER_VIEWS data dictionary views.

SELECT VIEW_NAME, TEXT
FROM USER-VIEWS
WHERE VIEW-NAME = "EMPLOYEE-VU".

4. Using your EMPLOYEES_VU view, enter a query to display all employees names and department.

SELECT Employee, department_id ;
FROM EMPLOYEES_VU;

5. Create a view named DEPT50 that contains the employee number, employee last names and department numbers for all employees in department 50.Label the view columns EMPNO, EMPLOYEE and DEPTNO. Do not allow an employee to be reassigned to another department through the view.

CREATE VIEW DEPT50 (Emp no, Employee, DeptNO) AS
SELECT employee-id, last-name, department-id
FROM Employees
WHERE department-id = 50 WITH CHECK OPTION CONSTRAINT DEPT-50 CHECK;

6. Display the structure and contents of the DEPT50 view.

DESCRIBE DEPT50
SELECT * FROM DEPT50;

7. Attempt to reassign Matos to department 80.

UPDATE Employees
SET department-id = 80 WHERE LAST_NAME = 'Matos" ;

8. Create a view called SALARY_VU based on the employee last names, department names, salaries, and salary grades for all employees. Use the Employees, DEPARTMENTS and JOB_GRADE tables. Label the column Employee, Department, salary, and Grade respectively.

CREATE VIEW salary_vu AS
SELECT e.last_name AS Employee, d.department_name AS Department
FROM Employees e
JOIN Departments d ON e.department_id = d.department_id
JOIN Job-grade ~~JOIN~~ J ON e.salary BETWEEN j.lowest_salary
AND j. highest_salary ;

| Evaluation Procedure | Marks awarded |
|---|---|
| Query(5) | 5 |
| Execution (5) | 5 |
| Viva(5) | 5 |
| Total (15) | 15 |
| Faculty Signature | 30/9/25 |